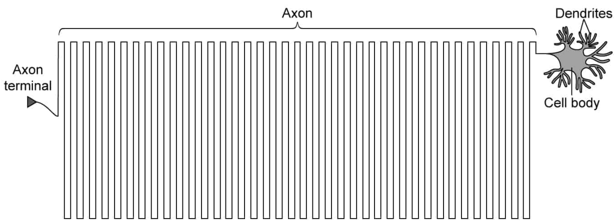


The above schematic represents a human motor neuron drawn to scale. Approximately 1% of the proteins found in axon terminal mitochondria are encoded by mitochondrial DNA. Given this, the mRNA transcripts for most mitochondrial proteins found in the axon terminal arise from what part(s) of a neuron?

I. Axon

II. Cell body (soma)

III. Dendrites

A. I only [12%]

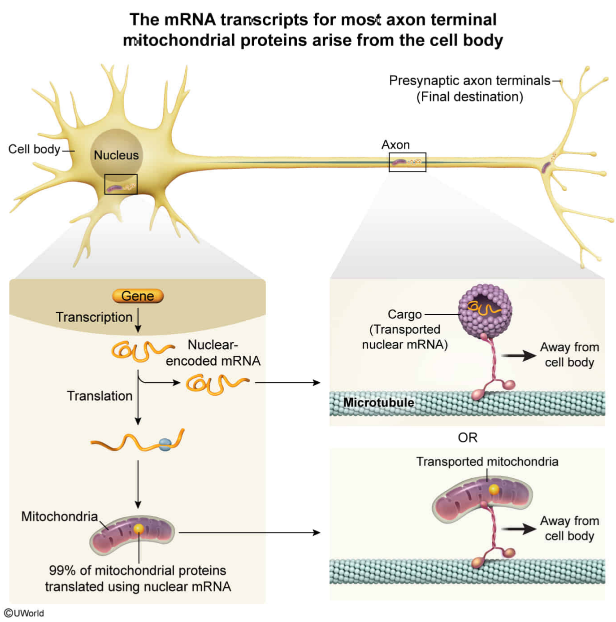
B. II only [62%]

C. I and II only [12%]

D. II and III only [12%]

Correct
62% answered correctly

Explanation:



Neurons are highly metabolically active. For example, neurons need to synthesize

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ATP to power the synthesis of proteins; for the transport of molecules, vesicles, and organelles (eg, mitochondria) within the neuron; and to maintain the ionic gradients necessary for impulse transmission along neurons.

This high demand for ATP is met predominantly via mitochondrial metabolism. However, the size and shape of neurons presents a challenge: Molecules and mitochondria necessary for axonal function must be supplied to the entire axon, which extends a relatively long distance from the nucleus. Neurons meet this challenge partly through the transport system that moves vesicles and mitochondria along axon terminals.

The question states that approximately 1% of the proteins found in axon terminal mitochondria are encoded by mitochondrial DNA and asks where the mRNA transcripts for most mitochondrial proteins found in the axon terminal arise. Because the vast majority (~99%) of axon terminal mitochondrial proteins are encoded by nuclear genes, the **mRNA for most axon terminal mitochondrial proteins arises from** the site for transcription of nuclear genes: **the nucleus**, which is **located in** the **neuron cell body (Number II)**.

(Numbers I and III) Because approximately 99% of the mitochondrial proteins found in axons are nuclear-encoded, the mRNA for these proteins must arise from the single site from which nuclear gene transcription occurs (ie, the nucleus). Therefore, the majority of transcripts for mitochondrial proteins likely arise from the nucleus (*not* the axon or dendrites).

Educational objective:
The neuron cell body contains the nucleus, the site where transcription of neuronal nuclear-encoded genes occurs.

Time spent:0 sec
Subject:
Biology

QID:402790
System:
Endocrine and Nervous Systems